One Blendshape Track, Any Avatar Rig: Real-Time Speech-Driven Faces in the Browser

98.1% of user-uploaded rigs with morph targets animate from one Audio2Face-3D track, with no per-rig retargeting step, at a median **3.8 s** from text to speaking face.

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Measured 2026-09-22 to 2026-11-04 on production traffic

1 Problem

Speech-driven facial animation ships as a **fixed blendshape vocabulary**, but the avatars people actually bring to a browser do not share one.

Audio2Face-3D emits 52 ARKit shapes plus tongue and head pose at 30 fps. On a platform where users upload, fork, auto-rig, and reconstruct their own characters, the receiving mesh exposes ARKit names, VRM vowels (`Aa Ih Ou Ee Oh`), Oculus visemes (`viseme_aa` …), vendor spellings, or no morph targets at all. In our sample, **54.9%** of files carry no facial morphs at all, and **11 of 211** of the morphed files expose no ARKit mouth shape under any spelling.

Goal. Drive every rig a user brings from the same track, in the browser, with no per-rig authoring, no model-side retargeting, and no frozen faces when coverage is partial.

Constraints. Audio must stay perceptually synchronized with the mouth; the client is a web page (no native runtime); rigs are untrusted glTF with meshopt and Draco compression, quantization, and arbitrary node structure.

3 Method: direct and derived mapping

At attach time the player inspects every morphed mesh once and binds each morph target to one of two paths. Per frame it does no string work: the track is sampled by the **audio element's clock** with linear interpolation, so the mouth cannot drift from the sound even when frames drop.

Direct path

A morph name is canonicalized (case, separators, an `ARKit_` prefix stripped) and, if it resolves to one of the 52 ARKit shapes, it is driven by that track column verbatim.

Derived path

A VRM vowel or Oculus viseme morph v with published ARKit composition c_v receives the weighted mean of the ARKit frame a over its components, the inverse of the authoring map:

```
w(v) = clamp01( Σk a[k] · cv[k]  /  Σk cv[k] )

// composition tables (excerpt)
viseme_aa: { jawOpen: 0.95 }
viseme_O:  { jawOpen: 0.50, mouthFunnel: 0.55 }
viseme_PP: { mouthClose: 0.90, mouthPressLeft: 0.40, mouthPressRight: 0.40 }
Ou (VRM): { mouthFunnel: 0.70, mouthPucker: 0.40, jawOpen: 0.20 }
```

Suppression rule

> A mesh that already has **any direct jaw or mouth shape** gets no derived bindings. Both paths at once double-drive the mouth (Section 6). Meshes with neither path fall back to amplitude lipsync from the playing audio, so a face is never frozen.

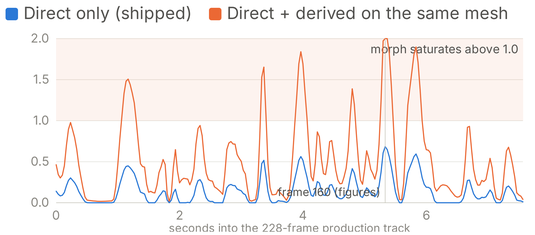


The same frame on three unrelated rigs. Frame 160 (t = 5.33 s, JawOpen = 0.68) of a production track for the sentence *"Hello, I am a three dimensional agent running in your browser, and my face is animated live by Audio2Face as I speak."*, applied by the shipping player to three avatar files with different authors, tools, and morph vocabularies. No rig-

4 Benchmarking method

- **Rig coverage.** 468 distinct avatar files from human accounts (at most 60 per creation source; seeded, QA, and synthetic accounts excluded). Each glTF's JSON chunk is fetched by HTTP range request and its morph dictionaries are handed to the *shipping* `A2FPlayer.attach()`, so the classification is the production code path, not a reimplementation.
- **Latency.** Ten consecutive production calls to the text-to-face endpoint with the same 22-word sentence, wall clock from request start to the full JSON response (TTS, resample, Audio2Face-3D stream). Calls refused by the per-client rate limit are excluded, not retried.
- **Overshoot.** Replay of the real 228-frame track through the composition tables, measuring the mapping-implied jaw opening with and without the suppression rule.

6 Ablation: why both paths cannot run at once



Mapping-implied jaw opening over the real track. With the suppression rule off, the ARKit `jawOpen` column and the visemes derived from it stack: the mouth saturates in **23%** of voiced frames (peak 6.1× the authored opening) instead of 0%.

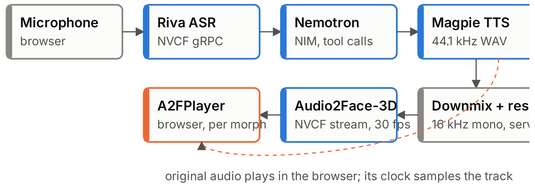


Same rig, same frame. **Derived only** is what a VRM or Oculus-style rig receives, produced here by hiding the rig's ARKit mouth shapes from the player: a usable mouth from visemes alone. **Both** is the failure the rule prevents.

7 Lessons and limitations

- **Name canonicalization beats allowlists.** One normalization function covered vendor spellings (`ARKit_jawOpen`, `jaw_open`, `JawOpen`) that an explicit rig list would have missed.
- **Clock the face from the audio element**, not from `requestAnimationFrame`. Drift was the first bug users noticed and the cheapest to remove.
- **Latency is dominated by upstream tails**, not by retargeting (sub-millisecond per frame). Streaming the first frames before TTS completes is the next step.
- **Limits.** Derived weights are a linear inverse of hand-authored compositions; expressive shapes with no vowel analogue (`mouthRollUpper`) are lost on viseme-only rigs. Head pose columns are applied only when a head bone exists. 54.9% of uploaded files have no facial morphs, which no mapping can fix.

2 Application of NVIDIA technology



End-to-end path for one spoken turn. Everything above the browser runs as hosted NVIDIA Cloud Functions over gRPC; the browser receives the original audio and a JSON track and does all retargeting locally.

- **Riva ASR** (Canary, multilingual) transcribes the user's microphone audio.
- **Nemotron** (`nemotron-3.5-lightning-30b-a3b` via NIM) produces the reply and tool calls.
- **Magpie TTS Multilingual** renders speech at 44.1 kHz; the server downmixes and linearly resamples a copy to 16 kHz mono for inference while the browser plays the original.
- **Audio2Face-3D** (NVCF bidirectional stream) returns per-frame weights for 55 shapes; the server packs them as `{fps, blendShapeNames, frames[{t, w[55]}]}`.

specific code or allowlist exists; new rigs are supported by adding a name composition, never a rig entry.

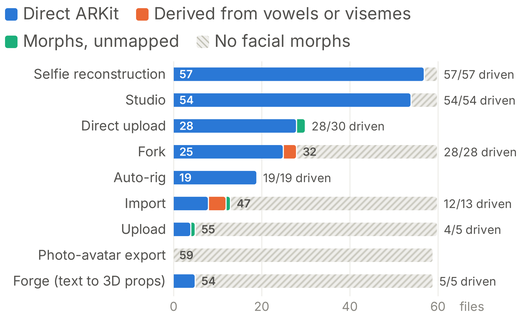
5 Results and outcomes

98.1%
of morphed rigs driven (207 of 211)

3.8 s
median text to animated speech, 7.3 s of audio

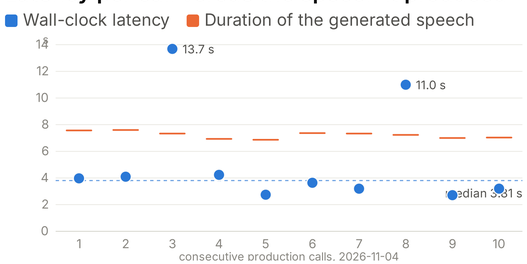
7
rigs rescued only by the derived path (4 still unmapped)

Which path drives each rig, by how the avatar was made



Sources that produce faces (selfie reconstruction, studio, auto-rig) are driven almost entirely by the direct path. The derived path matters where files are imported from other ecosystems (Import, Fork): 7 rigs that would otherwise have been frozen. Photo-avatar exports and text-to-3D props carry no facial morphs and get amplitude lipsync.

Latency per call versus the speech it produced



8 of 10 calls finished in 2.7 to 4.3 s for 6.9 to 7.6 s of speech (real-time ratio 0.39 to 0.61); two calls hit upstream tail latency at 11.0 and 13.7 s. Earlier runs: median 2.96 s (2026-10-08, six calls) and 1.29 to 2.19 s when audio is supplied directly and TTS is skipped (2026-09-22).

8 Reproduce

Source, data, scripts
Player, composition tables, coverage and latency scripts, and the raw measurements behind every figure.

Live demo
Speak or type; drop in your own glTF and watch which path binds each morph.

1. NVIDIA. Audio2Face-3D NIM, blendshape output and gRPC streaming. `docs.nvidia.com/ace/audio2face-3d-microservice`
2. NVIDIA. Riva ASR and Magpie TTS on NVIDIA Cloud Functions. `docs.nvidia.com/nim/riva`
3. Apple. ARFaceAnchor.BlendShapeLocation (52 shapes). `developer.apple.com/documentation/arkit/arfaceanchor/blendshapelocation`
4. VRM Consortium. VRM 1.0 expressions (aa, ih, ou, ee, oh). `github.com/vrm-c/vrm-specification`
5. Meta. Oculus Lipsync viseme reference (15 visemes). `developers.meta.com/horizon/documentation/unity/audio-ovrlipsync-viseme-reference`
6. Khronos. glTF 2.0, KHR_mesh_quantization, EXT_meshopt_compression. `github.com/KhronosGroup/glTF`
7. three.js r0.187.0, GLTFLoader and morph target animation. `threejs.org`